

ESPRIT Private School of engineering and technology



Academic Project

QoS Buddy

Prepared by

Khairi Hassine – Rayen Mannai – Majdi Guermazi – Eya Hamdi – Maher Hassen – Mariem Brik

Supervised by

Dorsaf Hriz & Fedi Baccar

Contents

List of Figures	2
Acronyms	3
General introduction	4
1 Project Setting	5
1.1 Introduction.....	5
1.2 Presenting the host organization	5
1.3 Presenting the project	5
1.3.1 Study and criticism of the existing.....	5
1.3.2 Problematic.....	7
1.3.3 Proposed Solution	7
1.4 Work method.....	8
1.5 Technical environment	8
1.6 Conclusion	10
2 Business Understanding	11
2.1 Introduction.....	11
2.2 Business objectives	11
2.3 KPIs	12
2.4 Data Science objectives	12
2.5 SDGs.....	13
2.6 Stakeholders.....	14
2.7 Conclusion	15
Bibliography	16

List of Figures

1	ESPRIT's Logo.....	5
2	Nagios logo.....	6
3	PRTG network monitor logo	6
4	Nokia logo.....	6
5	TDSP methodology.....	8
6	MLOps ecosystem.....	9
7	SDG 8.....	14
8	SDG 9.....	14

List of Acronyms

- **ELK:** Elasticsearch, Logstash, and Kibana
- **ICMP:** Internet Control Message Protocol
- **MTTD:** Mean Time to Detect
- **MTTR:** Mean Time to Resolution
- **NPM:** Network Performance Monitor
- **OPEX:** Operational Expenditure / Operational Costs
- **QoS:** Quality of Service
- **SLA:** Service Level Agreement
- **TDSP:** Team Data Science Process
- **XAI:** Explainable Artificial Intelligence

General introduction

In an era of rapidly evolving digital services, telecommunications networks play a critical role in supporting economic activity, social connectivity, and technological innovation. The exponential growth of data traffic, driven by cloud computing, video streaming, and emerging technologies such as 5G and IoT, has significantly increased the complexity of network management. In this context, ensuring a high Quality of Service has become a strategic priority for telecom operators, as network performance directly impacts user experience, operational efficiency, and contractual compliance.

Traditional network monitoring approaches are often reactive, relying on static thresholds and manual interventions that struggle to cope with the volume, velocity, and variability of modern network data. As a result, performance degradations and Service Level Agreement violations are frequently detected after they have already affected end users. To address these limitations, data-driven and intelligent solutions based on machine learning and artificial intelligence are increasingly adopted to enable proactive monitoring, early detection of anomalies, and informed decision-making.

This project focuses on **QoS Buddy**, an intelligent system designed to automatically monitor, analyze, and optimize Quality of Service in telecommunications networks. By continuously collecting and analyzing network performance indicators such as latency, jitter, packet loss, throughput, and availability, the solution aims to detect abnormal behaviors, forecast potential risks, and recommend corrective actions to maintain optimal network performance. Through the integration of machine learning techniques, QoS Buddy shifts network management from a reactive paradigm to a proactive and predictive approach.

At the core of this project lies the translation of business objectives—such as reducing SLA violations, improving network reliability, and minimizing operational costs—into well-defined Data Science problems. These problems encompass tasks such as time-series analysis, anomaly detection, classification, and regression, supported by appropriate evaluation metrics aligned with telecom performance requirements. The project emphasizes not only technical performance but also interpretability, ethical considerations, and responsible AI usage in a critical infrastructure context.

Chapter 1

Project Setting

1.1 Introduction

This chapter establishes the foundational context of the project. It begins with a presentation of the host organization, followed by an overview of the project through an analysis of the existing solution, its limitations, and the identified problem, leading to the proposed solution. The chapter also outlines the adopted work methodology and the technical environment, providing a clear framework for the rest of the project.

1.2 Presenting the host organization

ESPRIT (École Supérieure Privée d'Ingénierie et de Technologies) is a leading private higher education institution in Tunisia, renowned for its engineering programs accredited by the Tunisian Ministry of Higher Education. ESPRIT delivers cutting-edge training across various engineering specializations, including dedicated IT and Data Science programs.



Figure 1: ESPRIT's Logo

1.3 Presenting the project

This section presents the project by outlining the existing system, highlighting its limitations, and describing the proposed solution.

1.3.1 Study and criticism of the existing

In the field of network performance monitoring, several established commercial and open-source solutions provide visibility into network health, performance metrics, and fault

detection. Although these products are not inherently data-science driven, they represent the prevailing tools used in industry to maintain Quality of Service (QoS) and operational stability.



Figure 2: Nagios logo

Nagios is a widely used open-source network and infrastructure monitoring application that continuously checks the status of hosts, services, and network devices, alerting administrators when defined thresholds are crossed or when devices fail to respond. It uses standard protocols such as SNMP, SSH, WMI, and ICMP to gather performance data, and supports extensive customization through plugins, enabling monitoring of a large variety of network and system metrics. Nagios excels in flexible monitoring configurations but generally requires significant manual setup and lacks built-in predictive analytics capabilities.



Figure 3: PRTG network monitor logo

PRTG Network Monitor by Paessler is a comprehensive network monitoring platform that uses a sensor-based model to collect real-time data from network infrastructure, including routers, switches, servers, and applications. It supports automatic device discovery and a wide range of sensor types for performance metrics, such as CPU load, bandwidth usage, latency, and uptime, and provides customizable dashboards, alerts, and reporting interfaces. While PRTG simplifies monitoring setup and offers broad coverage of IT components, its focus remains primarily on descriptive analytics and threshold-based notifications rather than advanced data-driven forecasting.



Figure 4: Nokia logo

Nokia AI-Native Network Solutions are designed to support proactive monitoring and optimization of telecom networks in highly dynamic environments such as 5G and cloud-based infrastructures. Nokia integrates artificial intelligence and machine learning into its network management platforms to enable predictive traffic analysis, anomaly detection, and automated performance optimization. Through solutions such as Nokia Network Data Analytics Function (NWDAF) and its AI-native operations framework, the platform continuously analyzes QoS indicators, anticipates service degradations, and supports closed-loop automation for network assurance. While these solutions demonstrate strong predictive capabilities and advanced automation, they are proprietary, complex to deploy, and primarily targeted at large-scale operators, which can limit flexibility and transparency for customized data-driven experimentation.

1.3.2 Problematic

Modern telecommunication networks face escalating complexity due to increasing traffic volumes, heterogeneous services, and rising user expectations. While traditional monitoring tools such as Nagios and PRTG provide real-time visibility into key performance indicators (KPIs) like latency, bandwidth usage, and availability, and advanced solutions such as Nokia's AI-native platforms introduce predictive capabilities, current approaches remain either largely reactive or highly proprietary.

This operational model has the following limitations:

- **Reactive Detection:** Traditional tools like Nagios and PRTG often detect performance issues only after service degradation has already impacted end-user experience.
- **Static or Rigid Mechanisms:** Rule-based thresholds in conventional tools lack adaptability, while AI-native solutions rely on predefined closed-loop frameworks that limit customization and flexibility.
- **Data Complexity:** The massive volume and heterogeneity of network data challenge efficient, transparent analysis across both traditional monitoring systems and large-scale AI platforms.
- **Limited Accessible Predictive Intelligence:** While Nokia introduces predictive analytics, these capabilities are proprietary and opaque, whereas traditional tools lack advanced machine learning to identify hidden patterns and proactively prevent failures.

1.3.3 Proposed Solution

To address the identified limitations, we propose **QoS Buddy**, an intelligent decision-support system designed to transition network management from a reactive to a proactive paradigm. Leveraging machine learning and data science, the solution continuously monitors key performance indicators—such as latency, jitter, and packet loss—to detect anomalies before they impact end-users.

By integrating predictive analytics and unsupervised learning, QoS Buddy forecasts future risks and SLA violations, providing actionable optimization recommendations and explainable insights for network engineers. Ultimately, the system aims to automate complex decision-making, reduce operational costs (OPEX), and ensure a consistent, high-quality service experience in increasingly dynamic telecommunication environments.

1.4 Work method

To ensure the systematic development and deployment of the **QoS Buddy** solution, we have adopted the **Team Data Science Process (TDSP)**. Developed by Microsoft, TDSP is an agile, iterative data science methodology designed to improve collaboration and project efficiency through a structured lifecycle.

The project follows the five major stages of the TDSP lifecycle:

- **Business Understanding:** define objectives and identify data sources
- **Data Acquisition and Understanding:** ingest data and determine if it can answer the presenting question
- **Modeling:** feature engineering and model training
- **Deployment:** deploy into a production environment
- **Customer Acceptance:** customer validation if the system meets business needs

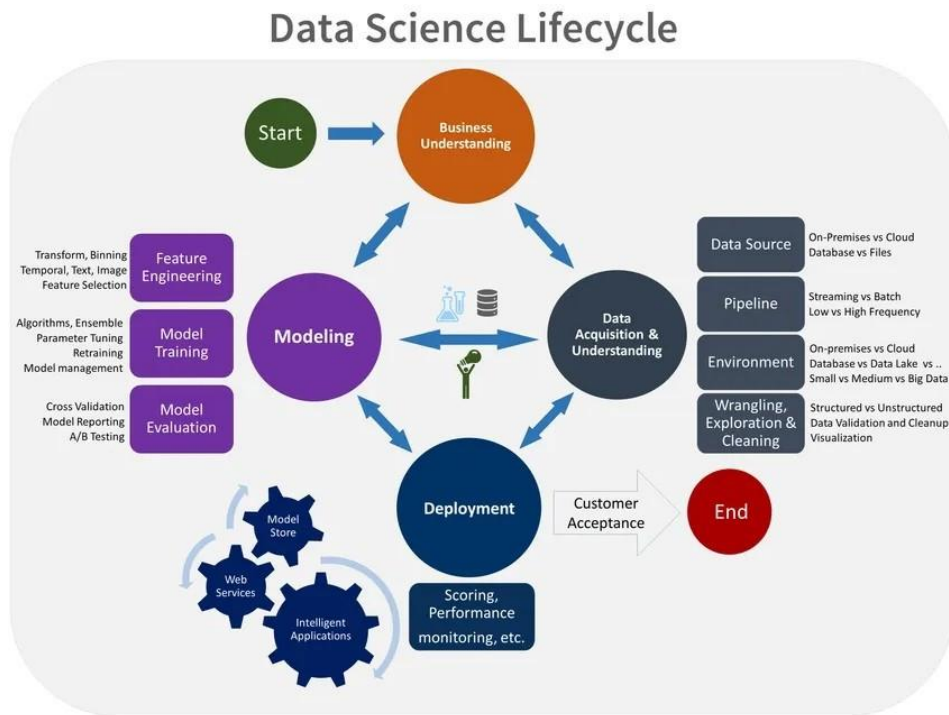


Figure 5: TDSP methodology

1.5 Technical environment

To implement the **QoS Buddy** solution and ensure its scalability, reliability, and reproducibility, we have established a robust **MLOps (Machine Learning Operations)** environment. This stack bridges the gap between model development and production-grade operations by automating workflows and ensuring continuous monitoring.

The following tools and technologies constitute our technical ecosystem:

- **Python:** The core programming language used for data manipulation, statistical analysis, and implementing machine learning algorithms via libraries such as Pandas, Scikit-learn, and TensorFlow.
- **FastAPI:** A modern, high-performance web framework used to build the inference API. It allows our machine learning models to be served as scalable RESTful services with automatic documentation.
- **Docker:** A containerization platform used to package the application, its dependencies, and the environment. This ensures that the system runs consistently across development, testing, and production stages.
- **Git:** The version control system used to track code changes, manage branches, and facilitate collaboration among team members.
- **Makefile:** Used to automate repetitive tasks such as environment setup, running tests, building Docker images, and triggering the ML pipeline with simple commands.
- **MLflow:** An open-source platform for the machine learning lifecycle. We use it for:
 - **Experiment Tracking:** Logging parameters, code versions, and metrics during model training.
 - **Model Registry:** Managing different versions of trained models and their deployment status.
- Elasticsearch & Kibana (ELK Stack):
 - **Elasticsearch:** A distributed search and analytics engine used to store and index real-time network logs and model prediction data.
 - **Kibana:** A visualization dashboard used to monitor network health and model performance in real-time through interactive charts and reports.

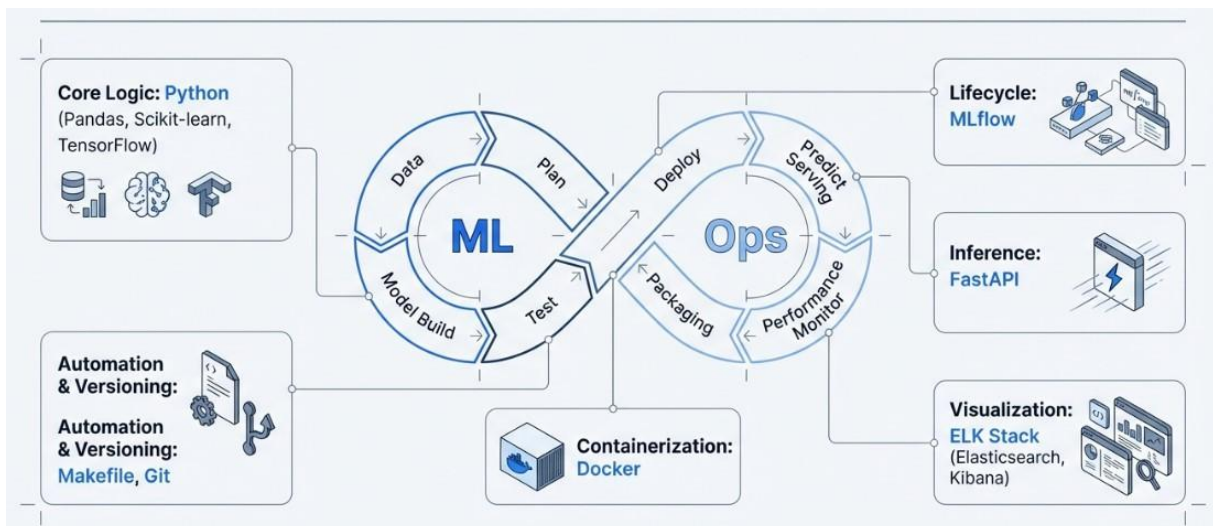


Figure 6: MLOps ecosystem

1.6 Conclusion

This chapter has established the foundational context of the project. It began by presenting the host organization, ESPRIT, and its commitment to engineering excellence. Next, an analysis of the existing network monitoring landscape highlighted the limitations of reactive, threshold-based tools, leading to the proposal of **QoS Buddy**—an intelligent, data-driven solution for proactive QoS management. Finally, the Team Data Science Process (TDSP) and a robust MLOps technical stack were introduced to provide a structured framework for development and deployment. This groundwork prepares for the upcoming phases of business understanding, data exploration and model implementation.

Chapter 2

Business Understanding

2.1 Introduction

This chapter focuses on establishing a clear understanding of the business context and objectives of the project. It identifies the key stakeholders involved, defines the business goals related to Quality of Service monitoring and optimization, and determines the success criteria through relevant KPIs. Furthermore, this chapter translates the identified business needs into concrete Data Science objectives, while taking into account sustainable development goals and ethical considerations. This structured business understanding provides the foundation for the subsequent phases of the TDSP methodology.

2.2 Business objectives

The objective of this project is to enhance the management and performance of telecom networks by leveraging data-driven techniques to monitor, predict, and optimize QoS. The business objectives are defined as follows:

- **BO1: Improve Network QoS:** Ensure consistently high network performance in terms of latency, throughput, jitter, and packet loss across all network zones in order to enhance the overall user experience. This objective focuses on continuously assessing network performance, identifying QoS degradations, and comparing current performance against expected standards and historical baselines.
- **BO2: Prevent SLA Violations Through Proactive Detection:** Anticipate service degradations early enough to prevent SLA breaches and reduce customer complaints. This objective aims to move from reactive network management to a proactive approach by detecting early warning signs of SLA violations, identifying hidden patterns preceding failures, and assessing the severity and potential impact of predicted incidents.
- **BO3: Automate and Optimize Network Operations (OPEX Reduction):** Reduce operational costs, manual interventions, and reaction time through intelligent and automated network management. This objective emphasizes the automation of corrective actions and decision-making processes, while ensuring transparency and explainability of automated recommendations for network engineers and managers.

2.3 KPIs

To evaluate the achievement of the defined business objectives, a set of measurable and operational KPIs is established. These indicators allow continuous monitoring of network performance, service reliability, and operational efficiency.

- **For BO1: Improve Network QoS:**

- **Average Network Latency (ms):** Measures the average time required for data packets to travel across the network.
- **Jitter (ms):** Evaluates the variability in packet delay, particularly critical for real-time services.
- **Packet Loss Rate (%):** Indicates the percentage of packets lost during transmission.
- **Throughput / Bandwidth Utilization (% or Mbps):** Assesses the effective data transmission rate compared to network capacity.
- **Network Availability / Uptime (%):** Measures the proportion of time the network remains fully operational.

- **For BO2: Prevent SLA Violations Through Proactive Detection:**

- **SLA Violation Rate (%):** Percentage of services or network segments that exceed SLA thresholds.
- **Mean Time to Detect Degradation (MTTD):** Average time required to detect QoS degradation before SLA breach. Prediction
- **Accuracy of SLA Risk (%):** Measures the correctness of SLA violation predictions generated by forecasting models.
- **False Alarm Rate (%):** Evaluates the number of incorrect alerts generated by early warning systems.

- **For BO3: Automate and Optimize Network Operations:**

- **Mean Time to Resolution (MTTR):** Average time required to resolve network incidents.
- **Manual Intervention Reduction (%):** Percentage decrease in human-operated troubleshooting tasks.
- **Automated Action Success Rate (%):** Measures the effectiveness of automated corrective actions.
- **Operational Cost Reduction (OPEX %):** Evaluates cost savings achieved through automation and optimization.

2.4 Data Science objectives

Based on the identified business objectives, this project defines a set of Data Science objectives aimed at transforming raw network data into actionable insights and automated decisions. These objectives guide the selection of analytical techniques, models, and evaluation metrics throughout the project.

- **For BO1:Improve Network QoS:**

- **DSO1.1: Real-Time QoS Estimation:** Develop predictive and descriptive models to estimate key QoS metrics—latency, jitter, packet loss, and throughput—per network segment in near real time.
- **DSO1.2: QoS Degradation Root Cause Analysis:** Apply multivariate statistical analysis and feature importance techniques to identify network components, traffic conditions, or configurations responsible for QoS degradations.
- **DSO1.3: Network Performance Benchmarking:** Perform comparative analysis between current QoS indicators and historical baselines or predefined performance standards to detect anomalies and deviations.

- **For BO2: Prevent SLA Violations Through Proactive Detection:**

- **DSO2.1: Early Warning System for SLA Breach Risk:** Build forecasting and classification models capable of predicting the probability of SLA violations within a predefined future time window.
- **DSO2.2: Hidden Pattern Discovery Before Failures:** Use unsupervised learning techniques (e.g., clustering, anomaly detection) to uncover latent patterns that typically precede network incidents.
- **DSO2.3: Impact Severity Classification:** Design classification models to categorize predicted incidents based on their expected user impact and SLA criticality.

- **For BO3: Automate and Optimize Network Operations:**

- **DSO3.1: Autonomous Network Adjustment Engine:** Develop optimization and recommendation models that suggest corrective actions such as traffic rerouting or resource reallocation.
- **DSO3.2: Explainable Decision Support and Reporting:** Implement explainable AI (XAI) techniques to generate interpretable insights and operational reports that justify automated decisions to network engineers and managers.

2.5 SDGs

This project contributes to several United Nations Sustainable Development Goals (SDGs) by promoting efficient, reliable, and responsible use of digital infrastructure through data-driven and intelligent network management.

SDG 8: Decent Work and Economic Growth:

- Improves the productivity of network engineers by reducing manual monitoring and repetitive operational tasks through intelligent automation.
- Supports economic activity by ensuring reliable network connectivity, which is essential for digital services and business operations.

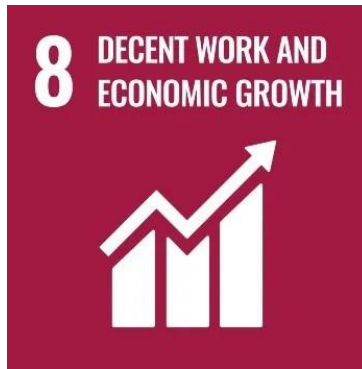


Figure 7: SDG 8

SDG 9: Industry, Innovation, and Infrastructure:

- Enhances the reliability and resilience of digital infrastructure by improving QoS and preventing service degradations.
- Supports the development of intelligent telecom networks through the integration of data science and automated decision-making mechanisms.



Figure 8: SDG 9

2.6 Stakeholders

The success of this project depends on the involvement and collaboration of several stakeholders with different roles and expectations:

- **Telecom Network Operators / Service Providers:** Primary stakeholders responsible for managing network infrastructure and ensuring service quality. They benefit from improved QoS, reduced SLA violations, and optimized operational costs.
- **Network Engineers and Operations Teams:** End users of the proposed solution. They rely on monitoring, alerts, and recommendations to detect issues, perform corrective actions, and maintain network performance.
- **Customers / End Users:** Indirect stakeholders affected by network performance. Improved QoS leads to better user experience, service reliability, and reduced service disruptions.
- **Business and Management Teams:** Use performance reports and KPIs to assess service quality, operational efficiency, and compliance with SLAs, supporting strategic and financial decision-making.

2.7 Conclusion

This chapter established a clear understanding of the business context of the project by identifying the key stakeholders, defining the business objectives, and determining the relevant performance indicators. The business needs were translated into precise Data Science objectives, ensuring alignment between operational goals and analytical solutions. Furthermore, the project's contribution to sustainable development goals was highlighted, confirming its relevance from both an economic and societal perspective. This business understanding phase provides a solid foundation for the subsequent data exploration and modeling stages.

Bibliography

- [1] PRTG network monitor. Paessler.
- [2] Nagios network monitoring. Nagios.
- [3] Nokia network Solutions. Nokia.
- [4] Team Data Science Process. Data Science PM.